Densenet

Fold1:
Train samples: 8012
Test samples: 2003
Fold 1 Train Distribution:

{'akiec': 261, 'bcc': 411, 'bkl': 880, 'df': 92, 'mel': 890, 'nv': 5364, 'vasc': 114}

Fold 1 Test Distribution:

{'akiec': 66, 'bcc': 103, 'bkl': 219, 'df': 23, 'mel': 223, 'nv': 1341, 'vasc': 28}

Accuracy Improved (0.8247628557164254 to 0.8347478781827259)

Evaluation report on Train data:

	precision	recall	f1-score	support
akiec	0.85	0.84	0.85	261
bcc	0.92	0.91	0.92	411
bkl	0.87	0.86	0.87	880
df	0.90	0.85	0.87	92
mel	0.82	0.75	0.78	890
nv	0.95	0.97	0.96	5364
vasc	0.95	0.94	0.94	114
accuracy			0.92	8012
macro avg	0.89	0.87	0.88	8012
weighted avg	0.92	0.92	0.92	8012

Evaluation report on Test data:

	precision	recall	f1-score	support
akiec	0.70	0.61	0.65	66
bcc	0.81	0.76	0.78	103
bkl	0.70	0.63	0.67	219
df	0.58	0.48	0.52	23
mel	0.62	0.58	0.60	223
nv	0.89	0.93	0.91	1341
vasc	0.92	0.82	0.87	28
accuracy			0.83	2003
macro avg	0.75	0.69	0.71	2003
weighted avg	0.83	0.83	0.83	2003

Epoch: 16 | train_loss: 0.2033 | train_acc: 0.9235 | test_loss: 0.5480 | test_acc: 0.8347

Fold 2:
Train samples: 8012
Test samples: 2003
Fold 2 Train Distribution:
{'akiec': 261, 'bcc': 412, 'bkl': 879, 'df': 92, 'mel': 890, 'nv': 5364, 'vasc': 114}
Fold 2 Test Distribution:
{'akiec': 66, 'bcc': 102, 'bkl': 220, 'df': 23, 'mel': 223, 'nv': 1341, 'vasc': 28}

Accuracy Improved (0.8102845731402896 to 0.8167748377433849), Saving the model.....

Evaluation report on Train data:

	precision	recall	f1-score	support
akiec	0.85	0.82	0.83	261
bcc	0.93	0.94	0.93	412
bkl	0.87	0.85	0.86	879
df	0.90	0.84	0.87	92
mel	0.82	0.77	0.79	890
nv	0.95	0.97	0.96	5364
vasc	0.96	0.95	0.95	114
accuracy			0.93	8012
macro avg	0.90	0.88	0.89	8012
weighted avg	0.92	0.93	0.92	8012

Evaluation report on Test data:

	precision	recall	f1-score	support
akiec	0.60	0.65	0.62	66
bcc	0.72	0.74	0.73	102
bkl	0.54	0.80	0.64	220
df	0.57	0.35	0.43	23
mel	0.66	0.39	0.49	223
nv	0.92	0.91	0.92	1341
vasc	0.96	0.82	0.88	28
accuracy			0.82	2003
macro avg	0.71	0.67	0.67	2003
weighted avg	0.83	0.82	0.81	2003

Epoch: 17 | train_loss: 0.1965 | train_acc: 0.9254 | test_loss: 0.6588 | test_acc: 0.8168 |

Fold3:
Train samples: 8012
Test samples: 2003
Fold 3 Train Distribution:
{'akiec': 262, 'bcc': 411, 'bkl': 879, 'df': 92, 'mel': 890, 'nv': 5364, 'vasc': 114}
Fold 3 Test Distribution:
{'akiec': 65, 'bcc': 103, 'bkl': 220, 'df': 23, 'mel': 223, 'nv': 1341, 'vasc': 28}
Accuracy Improved (0.799800299550674 to 0.8242636045931103)

Evaluation report on Train data:

	precision	recall	f1-score	support
akiec	0.72	0.68	0.70	262
bcc	0.87	0.87	0.87	411
bkl	0.75	0.72	0.73	879
df	0.75	0.64	0.69	92
mel	0.66	0.54	0.60	890
nv	0.91	0.95	0.93	5364
vasc	0.91	0.84	0.87	114
accuracy			0.86	8012
macro avg	0.79	0.75	0.77	8012
weighted avg	0.85	0.86	0.85	8012

Evaluation report on Test data:

	precision	recall	f1-score	support
akiec	0.53	0.63	0.57	65
bcc	0.76	0.65	0.70	103
bkl	0.67	0.62	0.65	220
df	0.43	0.43	0.43	23
mel	0.60	0.52	0.56	223
nv	0.90	0.94	0.92	1341
vasc	0.88	0.79	0.83	28
accuracy			0.82	2003
macro avg	0.68	0.65	0.67	2003
weighted avg	0.82	0.82	0.82	2003

Epoch: 10 | train_loss: 0.3817 | train_acc: 0.8587 | test_loss: 0.5106 | test_acc: 0.8243 |

Fold4:
Train samples: 8012
Test samples: 2003
Fold 4 Train Distribution:
{'akiec': 262, 'bcc': 411, 'bkl': 879, 'df': 92, 'mel': 891, 'nv': 5364, 'vasc': 113}
Fold 4 Test Distribution:
{'akiec': 65, 'bcc': 103, 'bkl': 220, 'df': 23, 'mel': 222, 'nv': 1341, 'vasc': 29}
Accuracy Improved (0.8192710933599601 to 0.8227658512231653), Saving the model.....

Evaluation report on Train data:

	precision	recall	f1-score	support
akiec	0.89	0.89	0.89	262
bcc	0.93	0.94	0.94	411
bkl	0.91	0.91	0.91	879
df	0.93	0.89	0.91	92
mel	0.87	0.84	0.85	891
nv	0.97	0.97	0.97	5364
vasc	0.97	0.96	0.97	113
accuracy			0.95	8012
macro avg	0.92	0.91	0.92	8012
weighted avg	0.95	0.95	0.95	8012

Evaluation report on Test data:

	precision	recall	f1-score	support
akiec	0.63	0.51	0.56	65
bcc	0.71	0.75	0.73	103
bkl	0.63	0.69	0.66	220
df	0.50	0.57	0.53	23
mel	0.61	0.58	0.59	222
nv	0.91	0.92	0.91	1341
vasc	0.89	0.55	0.68	29
accuracy			0.82	2003
macro avg	0.70	0.65	0.67	2003
weighted avg	0.82	0.82	0.82	2003

Epoch: 18 | train_loss: 0.1553 | train_acc: 0.9458 | test_loss: 0.6114 | test_acc: 0.8228 |

Fold5:
Train samples: 8012
Test samples: 2003
Fold 5 Train Distribution:
{'akiec': 262, 'bcc': 411, 'bkl': 879, 'df': 92, 'mel': 891, 'nv': 5364, 'vasc': 113}
Fold 5 Test Distribution:
{'akiec': 65, 'bcc': 103, 'bkl': 220, 'df': 23, 'mel': 222, 'nv': 1341, 'vasc': 29}
Accuracy Improved (0.8057913130304544 to 0.8062905641537693), Saving the model.....

Evaluation report on Train data:

	precision	recall	f1-score	support
akiec	0.81	0.79	0.80	262
bcc	0.93	0.94	0.94	411
bkl	0.83	0.83	0.83	879
df	0.80	0.68	0.74	92
mel	0.80	0.75	0.77	891
nv	0.95	0.96	0.96	5364
vasc	0.95	0.93	0.94	113
accuracy			0.91	8012
macro avg	0.87	0.84	0.85	8012
weighted avg	0.91	0.91	0.91	8012

Evaluation report on Test data:

	precision	recall	f1-score	support
akiec	0.44	0.54	0.49	65
bcc	0.72	0.67	0.69	103
bkl	0.59	0.70	0.64	220
df	0.48	0.70	0.57	23
mel	0.65	0.35	0.46	222
nv	0.89	0.92	0.91	1341
vasc	0.92	0.79	0.85	29
accuracy			0.81	2003
macro avg	0.67	0.67	0.66	2003
weighted avg	0.81	0.81	0.80	2003

Epoch: 17 | train_loss: 0.2274 | train_acc: 0.9145 | test_loss: 0.6495 | test_acc: 0.8063 |